Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/30/2006
PI Name:	Rafailovich, Miriam H. Ph.D.		
Project Title:	Microgravity Processing of Polymer Thin Films		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Polymers		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stony Brook	**State:**	NY
Zip Code:	11794-2275	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sokolov, J. (SUNY Stony Brook) Larson, D. (SUNY Stony Brook) Ulman, D. (Polytechnic University)		
Grant/Contract No.:	NAG8-1698		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bilayer films containing clays were examined and complete absence of dewetting was observed, This was taken as an indication that the large aspect ratio of the clay may pin the films at the interface. Study the factors involved in stabilizing a thin polymer films against dewetting on a surface. In particular the focus is on the effect of nano particle fillers in controlling dewetting and increasing mechanical stability on a nanometer scale. The goal this year was to determine the effect of aspect ratio on the dewetting characteristics. From the scanning force images we can determine the contact angles of the polymer coated on the different surfaces. We can then measure the interfacial energy between the coating and the surface. Four types of studies are undertaken: (a) Determination of the equilibrium morphology: These are samples that are annealed for very long times which tell us the state of lowest energy and allow us to apply equilibrium theories to interpretation of the results. (b) Study of the dewetting dynamics: By studying the time development of the dewetting process we can infer the effective viscosity of the polymer on the given surface use various liquid flow theories. Furthermore, we can determine the effect of confinement on the polymer structure and dynamics. (c) Study the effect of nanoparticle fillers as a function of aspect ratio and functional group/polymer interactions. (d) Determine the effect of fillers on the mechanical strength of polymers as a function of the relative dimensions of the filler/polymer chain/ and feature size.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)